

Macroeconomics I — MPE Insper

Mock Final Exam 1 (Simulado 1)

Instructions. Four blocks of 25 points each (100 points). Suggested time: 3 hours. Multiple-choice items: mark the single correct option (no penalty for wrong answers). *True/False* items: every 2 wrong items cancel 1 right item (score = $\max\{0, \text{right} - \text{wrong}/2\} \times \text{item value}$); blank items do not count as wrong. Discursive questions: show every step; box your final answers. Numerical values are illustrative. Notation follows Kurlat (2020).

Block I — Measurement

(Lecture 1; Kurlat chs. 1–2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) An economy produces two goods. In Year 1, good X costs 10 and 10 units are produced; good Z costs 5 and 20 units are produced. In Year 2, X costs 5 and 20 units are produced; Z costs 10 and 16 units are produced. Real GDP growth from Year 1 to Year 2 measured at Year-1 prices and at Year-2 prices is, respectively:

- A. 40% at Year-1 prices; 4% at Year-2 prices.
- B. 4% at Year-1 prices; 40% at Year-2 prices.
- C. 30% at Year-1 prices; 30% at Year-2 prices.
- D. 40% at Year-1 prices; 30% at Year-2 prices.
- E. 20.7% at Year-1 prices; 20.7% at Year-2 prices.

b) (3 points) Why is the GDP of a low-income country, converted into dollars at the market exchange rate, systematically *lower* than the same GDP converted at purchasing power parity (PPP)?

- A. Tariffs and transport costs keep traded-goods prices apart across countries, and the market rate cannot close that gap.
- B. Non-traded goods and services are cheap where wages are low, and the market rate, set by traded goods, ignores this.
- C. The market rate is driven by capital flows rather than by trade, so it undervalues every poor-country currency by the same factor.
- D. Large informal sectors are left out of the official statistics, so a big share of production is missing from measured GDP.
- E. Quotas and import licences raise traded-goods prices in poor countries, which inflates their measured price level.

c) (2 points, T or F) In a two-good economy, if between two years quantities shift toward the good whose relative price fell, then real GDP growth measured at initial-year prices exceeds real GDP growth measured at final-year prices.

d) (2 points, T or F) Brazil imports all the smartphones it consumes and produces none. A rise in their dollar price raises the IPCA and the GDP deflator by the same proportion, since both measure “the price level”.

Question 2. (15 points) IBGE and the smartphone boom. Consider a stylised Brazil producing only soybeans (s) and smartphones (m). Values in billions of reais; population in millions.

	p_s	q_s	p_m	q_m	Population
2015	10	60	20	20	200
2025	12	60	12	40	210

a) (5 points) Compute nominal GDP in both years; real GDP in 2015 prices and in 2025 prices; the real growth rate between 2015 and 2025 in each base; and the Fisher (chained) growth rate. State which year's value is in the denominator of each rate.

Solution:

b) (5 points) The two bases give different answers. For each one, say whether it overstates or understates growth relative to the Fisher index, and explain *why*, using the movement of relative prices and quantities in the table.

Solution:

c) (5 points) Compute real GDP growth per capita in each base and with Fisher. Which number, aggregate or per capita, answers “did the average Brazilian become better off?” and which answers “how much bigger is the Brazilian economy?” What does neither answer?

Solution:

Block II — Growth and the Solow model

(Lectures 2–3; Kurlat chs. 3–5 • 25 points)

Question 3. (10 points) Objective items. Items (a)–(b) use the Solow model in per-worker terms, $y = Ak^\alpha$, with no technological progress, $A = 1$, $\alpha = 1/3$, $s = 0.24$, $n = 0.01$, $\delta = 0.05$, and $(1 + n)k_{t+1} = (1 - \delta)k_t + sAk_t^\alpha$.

a) (3 points) Steady-state capital per worker k^* is:

- A. 2
- B. 4
- C. 8
- D. 10.52
- E. 64

b) (3 points) About the Golden-Rule saving rate s_{gold} of this economy:

- A. $s_{gold} = 0.24$; the economy is already at the Golden Rule and c^* cannot rise.
- B. $s_{gold} = 1$; saving all output maximises k^* and therefore also maximises c^* .
- C. $s_{gold} = 2/3$; it equals the labour share, so raising s to $2/3$ maximises c^* .
- D. $s_{gold} = 1/3$; raising s from 0.24 toward it raises c^* in the long run.
- E. $s_{gold} = 1/3$; raising s from 0.24 lowers c^* , as the economy is dynamically inefficient.

c) (2 points, T or F) In the Solow model without technological progress, a permanently higher saving rate raises the long-run growth rate of output per worker.

d) (2 points, T or F) Starting from a steady state, a permanent fall in population growth n raises k^* and temporarily raises the growth rate of output per worker above its steady-state rate.

Question 4. (15 points) German reunification, 1990. Treat West (W) and East (E) Germany as two closed Solow economies without technological progress, $Y_i = A_i K_i^\alpha L_i^{1-\alpha}$ with a common $\alpha = 1/2$ and $\delta = 0.05$, and $(1 + n_i)k_{i,t+1} = (1 - \delta)k_{i,t} + s_i A_i k_{i,t}^\alpha$, where $k = K/L$ and $y = Y/L$ (per worker = per capita). Units: L in millions, K in index units.

	A_i	s_i	n_i	$L_{i,0}$	$K_{i,0}$
West	2	0.275	0	64	7744
East	1	0.375	0.025	16	256

a) (5 points) Compute k and y in each region in period 0, and each region's steady state k_i^* , y_i^* . What does the model predict about growth in each region, and about the gap between them? Name the prediction.

Solution:

b) (5 points) Unification: from 1990 all production uses West technology $A_W = 2$ with the pooled capital and labour of both regions. Compute the new y and its change for West and for East workers (in levels and in % of the pre-unification value), and the change in aggregate German output Y . Explain *where* each region's gain or loss comes from.

Solution:

c) (5 points) After unification Germany saves at the West's rate s_W and its labour force grows at the labour-weighted average of n_W and n_E . Compute k_U^* and y_U^* . Is unified Germany below its steady state? Compare y_U^* with y_W^* and explain the difference.

Solution:

Block III — Microfoundations

(Lectures 4–6; Kurlat chs. 6, 7, 9 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) A household has one unit of time, preferences $\ln c + \psi \ln l$ with $\psi = 1$, wage $w = 10$, pays a tax τ on labour income and receives a lump-sum transfer $T = 2$, so $c = (1 - \tau)w(1 - l) + T$. The labour-income tax rises from $\tau = 0.2$ to $\tau = 0.5$ (T unchanged). Hours:

- A. fall from 0.375 to 0.300: leisure became cheaper, an effect that would lower hours even with $T = 0$.
- B. stay at 0.500: with log utility the income and substitution effects cancel whatever T is.
- C. rise from 0.375 to 0.450: the lower net wage makes the household poorer, so it works more.
- D. fall from 0.500 to 0.375: the tax lowers the price of leisure one-for-one with τ .
- E. fall from 0.375 to 0.300: the fixed T grows relative to net earnings; with $T = 0$ hours would not move.

b) (3 points) In a search economy with $m(V, U) = \mu V^\alpha U^{1-\alpha}$, one firm posts one extra vacancy. Which statement is correct?

- A. The posting firm and the other firms gain; unemployed workers lose, as matches per worker fall.
- B. Unemployed workers gain (f rises); other firms lose (q falls); the posting firm gains privately.
- C. The posting firm loses, because its own q falls; this private cost is the congestion externality.
- D. Everyone gains, since m rises one-for-one with V , so the extra vacancy creates no externality.
- E. Unemployed workers lose (f falls); other firms gain (q rises); the posting firm is unaffected.

c) (2 points, T or F) A fall in matching efficiency μ , which raises the vacancy rate needed at every unemployment rate, is a movement along the Beveridge curve.

d) (2 points, T or F) In the general-equilibrium economy with a fixed capital stock, no investment, and preferences $\ln c + \psi \ln l$, a rise in TFP A leaves hours worked unchanged.

Question 6. (15 points) A temporary tax cut financed by public debt. The Brazilian government considers cutting this year's income tax and repaying the resulting debt next period. A household lives two periods and solves

$$\max_{c_1, c_2, a} u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad c_1 + a = y_1 - \tau_1, \quad c_2 = y_2 - \tau_2 + (1 + r)a,$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ ($u = \ln c$ if $\sigma = 1$), $\beta \in (0, 1)$, $\sigma > 0$. Baseline values: $\sigma = 1$, $\beta = 0.8$, $r = 0.25$ (a period is roughly a decade), $y_1 = 60$, $y_2 = 150$, $\tau_1 = \tau_2 = 20$ (thousand reais).

a) (5 points) By substituting the constraints into the objective (no Lagrangian needed), derive the Euler equation and solve for c_1 , c_2 and a in closed form. Evaluate them at the baseline.

Solution:

b) (5 points) Set $y_2 = \tau_1 = \tau_2 = 0$ (general σ). Compute $\partial c_1 / \partial r$, isolate the factor that determines its sign, and state, in words consistent with that sign, what happens to c_1 when r rises for $\sigma > 1$, $\sigma = 1$ and $\sigma < 1$.

Solution:

c) (5 points) Return to the baseline. The government cuts τ_1 by $\Delta = 10$ and raises τ_2 by $(1+r)\Delta$. (i) Find the effect on c_1 , c_2 and a . Does the timing of taxes matter? (ii) Now the household faces a borrowing limit $a \geq -b$ with $b = 20$. Find c_1 , c_2 , a before and after the same tax change. (iii) Give two concrete examples of Brazilian households for whom such a limit is likely to bind.

Solution:

Block IV — Money, inflation and AD–AS

(Lectures 7–9; Kurlat chs. 10–11; Benigno (2015) • 25 points)

Question 7. (10 points) Objective items.

a) (3 points) About the quantity equation $MV = PY$:

- A. It is a theory of money demand: with Y fixed, doubling M doubles P at every horizon.
- B. It is an identity which proves neutrality, because V and Y are constant by definition.
- C. It is a behavioural equation: V is a structural parameter that the central bank sets.
- D. It is an identity, since $V \equiv PY/M$; causation needs assumptions on V and on what is exogenous.
- E. It is an identity which proves causation runs from P to M , because firms set prices first.

b) (3 points) In steady state with constant i , $\pi = \mu - \eta g$. The central bank targets $\pi^* = 3\%$ with $g = 2\%$ and sets μ assuming $\eta = 1$ (Cambridge). Money demand is actually Baumol–Tobin. Inflation is:

- A. 2%
- B. 3%
- C. 4%
- D. 5%
- E. 6%

c) (2 points, T or F) When the central bank targets the nominal interest rate, the money-market equilibrium $M = pm^D(Y, i)$ determines the quantity of money.

d) (2 points, T or F) In Benigno’s AD–AS model, a temporary rise in firms’ desired mark-up shifts the AD curve to the left.

Question 8. (15 points) Pix and the demand for money. Since 2020 Pix, Brazil’s instant-payment system, has lowered the cost of turning deposits into means of payment. Model real money demand with Baumol–Tobin, $m^D = \sqrt{\frac{FY}{2i}}$, where F is the real cost of each conversion. Assume real output grows at $g = 2\%$, F falls at the constant rate $\dot{F}/F = f = -4\%$, and i is constant. The money market clears: $M = pm^D(Y, i, F)$.

a) (5 points) Derive velocity V implied by this model. Explain why V is not constant, and compute its growth rate under the assumptions above. Interpret velocity in words.

Solution:

b) (5 points) By totally differentiating $M = pm^D(Y, i, F)$ with respect to time, show that in steady state $\pi = \mu - \frac{1}{2}(g + f)$. Before Pix ($f = 0$) the BCB grew money at the rate that delivered $\pi = 3\%$. Which money growth μ delivers 3% after Pix? What inflation results if the BCB keeps the old μ ? Explain the mechanism in words.

Solution:

c) (5 points) Suppose instead the BCB sets the Selic (an interest-rate target i^*) and lets the money supply be whatever it must be. When Pix lowers F , what adjusts in the money market in the short run (prices sticky)? Is the Pix effect inflationary under this regime in the long run? Contrast with the money-growth rule of (b).

Solution: